Hybrid training under residual misspecification

Results by dataset × misspec family × γ × method

Exact-greedy CTR (%), paired contrasts, and DiD with 95% t CIs

OPC project

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Abstract

This note explains the experiment from first principles and then reports **every locked cell with numbers**: absolute CTR levels, method contrasts (hybrid / OPC vs NLL), and difference-in-differences (DiD) with 95% confidence intervals. Tables are stratified by **dataset**, **misspec family**, γ, and **method**. The full hybrid grid (`nlres_grid_*`, n=40, methods including $f \in \{0.25, 0.5, 0.75\}$ and adapt) is still running for non-ML catalogs; ML SoftAND / SoftThresh / $R^\star$ grids and earlier multi-dataset $R^\star$ locks are included.

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1 Problem setup (for a reader not in the loop)

1.1 Offline policy learning with logged bandit feedback

We have users x, items / actions $a \in \mathcal{A}$, and binary rewards $r \in \{0, 1\}$ (click). A logging policy π_0 collected a dataset $\mathcal{D} = \{(x_i, a_i, r_i, p_i)\}$ with propensities $p_i = \pi_0(a_i \mid x_i)$. A learner outputs a new policy π_θ. We evaluate with the **exact-greedy CTR** of π_θ in a simulated world whose click probabilities $q(x, a) = \Pr(r{=}1 \mid x, a)$ are known to the evaluator (but not used as training labels for the policy, except through the usual reward model / IPS machinery of each method).

Reported metric (percentage points of CTR):

$$V(\pi) = 100 \cdot \mathbb{E}_{x\sim\nu}\Big[q\big(x, \arg\max_a \pi(a \mid x)\big)\Big]. \tag{1}$$

1.2 Ground-truth world and the misspecification dial γ

The environment logit is an affine-bilinear "clean" score plus a residual:

$$\ell_\gamma(x, a) = \underbrace{b + u_x^\top v_a}_{\text{GT bilinear}} + \gamma \cdot \alpha \cdot \phi(x, a), \qquad \mathrm{sd}(\phi) = 1, \tag{2}$$

and $q = \sigma(\ell_\gamma)$ after a global offset that freezes a target reference CTR (here 5% under the logger). The scalar $\gamma \geq 0$ is the **dial**: $\gamma = 0$ is well-specified bilinear truth; large γ makes the residual ϕ dominate ranking.

Three residual families ϕ are used (unit-SD by construction after normalization):

- $R^\star$ (even map on a second random projection; $s{=}1, m{=}4, \psi{=}\tanh$). Empirically $\mathrm{sd}(\gamma\alpha\phi)/\mathrm{sd}(\mathrm{GT}) \approx 2.4\gamma$ (overshoots the nominal dial).
- **SoftAND*** (multiplicative gate; tracks sd_ratio $\approx \gamma$).
- **SoftThresh**τ=1.5 (thresholded residual; also tracks sd_ratio $\approx \gamma$).

A useful summary of "how loud is ϕ" is the logit variance share $\mathrm{Var}(\gamma\alpha\phi)/\mathrm{Var}(\ell_\gamma)$. SoftAND / SoftThresh at γ=0.5 keep $\approx 20\%$ variance on ϕ; $R^\star$ at the same nominal γ is louder ($\approx 59\%$).

1.3 Learners

All primary learners are matched linear policies with Optuna-tuned LR / epochs ($n_{\text{train}}{=}10^5$, $n_{\text{val}}{=}2{\cdot}10^4$, 20 trials, bias cell `high`). Methods in the tables:

- **NLL**: maximum likelihood on logged clicks (no IPS correction).
- **OPC/DR**: doubly robust offline policy learning (direct method + IPS).
- **hybrid_seq_**f: train with NLL for the first fraction f of the (selected) epoch budget, then switch to DR for the remainder. Locked default $f^\star{=}0.5$; the full grid also reports $f \in \{0.25, 0.75\}$.
- **hybrid_seq_adapt**: start with NLL, switch to DR when validation NLL plateaus (patience 2), with switch fraction clipped to $[0.25, 0.75]$.

1.4 Estimands: levels, contrasts, DiD

For a fixed dataset, misspec family, and γ, write $V_m(\gamma)$ for method m. Paired over seeds $s = 1, \ldots, n$:

$$\Delta_{m,m'}(\gamma) = V_m(\gamma) - V_{m'}(\gamma), \tag{3}$$

$$\mathrm{DiD}_{m,m'}(\gamma_1; \gamma_0) = \Delta_{m,m'}(\gamma_1) - \Delta_{m,m'}(\gamma_0). \tag{4}$$

Primary contrast vs likelihood: $\Delta_{\mathrm{H,NLL}}$ with H = `hybrid_seq_f0.5`. DiD asks: *does the hybrid advantage over NLL grow when we turn up misspecification?* CIs are seed-level Student-t: $\bar{x} \pm t_{0.975,n-1}\,\mathrm{SE}$, with $\mathrm{SE} = s/\sqrt{n}$.

1.5 Decision regret (no γ oracle)

Across locked cells, cell regret of a rule that picks hybrid or NLL is the forgone $|\Delta_{\mathrm{H,NLL}}|$ when the pick disagrees with the sign of the estimated mean. Empirically, **always-hybrid** has far smaller summed regret than always-NLL when γ is unknown.

2 Status snapshot

- **ML full hybrid grid** (`nlres_grid_ml_*`): SoftAND complete across γ; SoftThresh / $R^\star$ largely complete or near-complete (n up to 40; some mid-cells still filling $f \in \{0.25, 0.75, \text{adapt}\}$).
- **Anime** grid: in flight (partial seeds).
- **Myket / KuaiRand / KuaiRec** full hybrid grids: queued; earlier Phase-1 $R^\star$ locks (n=20) are included below.
- SoftAND n=40 catchup on ML (seeds 320–339) is complete for prior tags.

3 When to use hybrid (aggregate)

Prefer hybrid_seq_f0.5 as the default vs pure likelihood when residual misspecification strength is unknown. Across dataset×misspec cells, summed regret of always-NLL is much larger than always-hybrid: well-spec (γ=0) costs are typically ≤0.5–0.8pt, while mid/extreme ϕ misspec yields +0.5 to +6pt H–NLL with CI LB¿0 on most R* and extreme SoftAND/SoftThresh cells. Use pure NLL only when you have a strong prior of bilinear well-specification and need to squeeze the last γ=0 points (e.g. kuairand full-catalog γ=0). OPC–NLL≥0.5 is a conservative 'misspec is loud' flag but is not a good gate for switching away from hybrid—it misses many mid-γ hybrid wins where OPC still trails NLL.

Best regret rule. `hybrid_unless_OPC_NLL_le_-1.0`. Summed regret (pt): always-NLL 74.65, always-hybrid 3.52, rule 2.95. Sign agreement with H−NLL: 41; clear wins caught: 26/28.

4 Dial calibration (MovieLens reward ratios)

Uniform (x, a) pairs; designed sd(ϕ) = 1 so SoftAND/SoftThresh track sd_ratio ≈ γ. $R^\star$ overshoots (≈ 2.4×).

family	γ	sd_ratio	var_share$_\phi$	reward_ratio	CTR lift$_\phi$
rstar	0.00	0.000	0.0%	1.00	+0.00
rstar	0.25	0.597	26.3%	1.15	+0.00
rstar	0.50	1.194	58.7%	1.72	+0.01
rstar	0.75	1.791	76.2%	3.14	+0.02
rstar	1.00	2.388	85.1%	6.54	+0.03
rstar	2.00	4.776	95.8%	209.86	+0.04
rstar	4.00	9.551	98.9%	4.08e+05	+0.04
rstar	10.00	23.878	99.8%	4.47e+10	+0.04
rstar	100.00	238.776	100.0%	4.70e+10	+0.05
softand	0.00	0.000	0.0%	1.00	+0.00
softand	0.25	0.249	5.8%	1.04	+0.00
softand	0.50	0.498	19.9%	1.14	+0.00
softand	0.75	0.747	35.8%	1.27	+0.01
softand	1.00	0.996	49.8%	1.44	+0.01
softand	2.00	1.992	79.9%	2.43	+0.02
softand	4.00	3.984	94.1%	8.28	+0.03
softand	10.00	9.959	99.0%	896.35	+0.04
softand	100.00	99.592	100.0%	4.25e+10	+0.04
softthresh	0.00	0.000	0.0%	1.00	+0.00
softthresh	0.25	0.250	5.9%	1.04	+0.00
softthresh	0.50	0.500	20.0%	1.17	+0.00
softthresh	0.75	0.750	36.0%	1.35	+0.01
softthresh	1.00	1.001	50.0%	1.57	+0.01
softthresh	2.00	2.001	80.0%	3.16	+0.02
softthresh	4.00	4.003	94.1%	19.64	+0.03
softthresh	10.00	10.006	99.0%	2.41e+04	+0.04
softthresh	100.00	100.065	100.0%	4.25e+10	+0.04

5 Compact DiD index (primary contrasts, $n \geq 10$)

dataset	misspec	γ_0	γ_1	contrast	n	DiD	95% CI
anime	R*	0	0.6	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.10	[-0.26, +0.46]
anime	R*	0	0.6	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.50	[+0.08, +0.92]
kuairand	R*-full	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.17	[-0.28, +0.62]
kuairand	R*-full	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.61	[+0.12, +1.11]
kuairand	R*-mlsize	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+1.18	[+0.69, +1.67]
kuairand	R*-mlsize	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+1.31	[+0.92, +1.69]
kuairec	R*-full	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+1.56	[+0.92, +2.20]
kuairec	R*-full	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+1.30	[+0.72, +1.88]
kuairec	R*-mlsize	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.94	[+0.46, +1.42]
kuairec	R*-mlsize	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.72	[-0.06, +1.50]
ml	R*	0	0.25	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.26	[+0.03, +0.49]
ml	R*	0	0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.36	[+0.12, +0.61]
ml	R*	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.12	[+0.78, +1.47]
ml	R*	0	0.5	$V_{f0.25} - V_{\mathrm{NLL}}$	10	+1.82	[+0.58, +3.05]
ml	R*	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.11	[+0.72, +1.50]
ml	R*	0	0.5	$V_{f0.75} - V_{\mathrm{NLL}}$	10	+1.71	[+1.02, +2.39]
ml	R*	0	0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+2.01	[+1.59, +2.42]
ml	R*	0	0.75	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+1.75	[+1.20, +2.31]
ml	R*	0	0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.73	[+1.23, +2.24]
ml	R*	0	0.75	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+1.67	[+1.28, +2.05]
ml	R*	0	0.75	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+1.76	[+1.37, +2.15]
ml	R*	0	1	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+2.66	[+2.22, +3.09]
ml	R*	0	1	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+2.45	[+1.90, +3.01]
ml	R*	0	1	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+2.28	[+1.78, +2.77]
ml	R*	0	1	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+2.17	[+1.69, +2.64]
ml	R*	0	1	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+2.25	[+1.74, +2.76]
ml	R*	0	2	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+4.46	[+3.96, +4.96]
ml	R*	0	2	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+3.48	[+2.69, +4.27]
ml	R*	0	2	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.63	[+2.95, +4.31]
ml	R*	0	2	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+3.33	[+2.68, +3.98]
ml	R*	0	2	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+3.28	[+2.69, +3.88]
ml	R*	0	4	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+5.56	[+5.02, +6.09]
ml	R*	0	4	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+4.32	[+3.46, +5.18]
ml	R*	0	4	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+4.22	[+3.46, +4.98]
ml	R*	0	4	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+4.19	[+3.44, +4.94]
ml	R*	0	4	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+4.15	[+3.39, +4.91]
ml	R*	0	10	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	36	+6.24	[+5.67, +6.82]
ml	R*	0	10	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+4.72	[+3.88, +5.57]
ml	R*	0	10	$V_{f0.5} - V_{\mathrm{NLL}}$	36	+4.84	[+3.93, +5.74]
ml	R*	0	10	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+4.49	[+3.66, +5.32]
ml	R*	0	10	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+4.39	[+3.62, +5.15]
ml	R*	0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	36	+6.55	[+5.89, +7.21]
ml	R*	0	100	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+4.81	[+3.95, +5.68]
ml	R*	0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	36	+4.76	[+3.95, +5.56]
ml	R*	0	100	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+4.44	[+3.66, +5.23]
ml	R*	0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+4.36	[+3.51, +5.21]
ml	R*-adapt	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+1.35	[+0.88, +1.83]
ml	R*-adapt	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+1.19	[+0.59, +1.79]
ml	R*-adapt	0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+1.44	[+1.03, +1.85]
ml	SoftAND*	0	0.25	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.00	[-0.19, +0.19]
ml	SoftAND*	0	0.25	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.26	[+0.06, +0.46]
ml	SoftAND*	0	0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.09	[-0.11, +0.28]
ml	SoftAND*	0	0.25	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.19	[-0.03, +0.41]
ml	SoftAND*	0	0.25	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.22	[-0.00, +0.45]
ml	SoftAND*	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.13	[-0.08, +0.34]
ml	SoftAND*	0	0.5	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.39	[+0.15, +0.63]
ml	SoftAND*	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.35	[+0.13, +0.58]
ml	SoftAND*	0	0.5	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.42	[+0.15, +0.68]
ml	SoftAND*	0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.38	[+0.11, +0.65]
ml	SoftAND*	0	0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.35	[+0.10, +0.60]
ml	SoftAND*	0	0.75	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.71	[+0.41, +1.02]
ml	SoftAND*	0	0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.55	[+0.23, +0.86]

dataset	misspec	γ_0	γ_1	contrast	n	DiD	95% CI
ml	SoftAND*	0	0.75	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.70	[+0.38, +1.03]
ml	SoftAND*	0	0.75	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.76	[+0.43, +1.09]
ml	SoftAND*	0	1	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.47	[+0.21, +0.73]
ml	SoftAND*	0	1	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.73	[+0.34, +1.12]
ml	SoftAND*	0	1	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.82	[+0.52, +1.13]
ml	SoftAND*	0	1	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.82	[+0.46, +1.18]
ml	SoftAND*	0	1	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.81	[+0.47, +1.15]
ml	SoftAND*	0	2	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.05	[+0.72, +1.38]
ml	SoftAND*	0	2	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+1.47	[+0.88, +2.06]
ml	SoftAND*	0	2	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.71	[+1.24, +2.18]
ml	SoftAND*	0	2	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+1.52	[+1.15, +1.90]
ml	SoftAND*	0	2	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+1.53	[+1.10, +1.96]
ml	SoftAND*	0	4	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.88	[+1.43, +2.32]
ml	SoftAND*	0	4	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+2.43	[+1.77, +3.09]
ml	SoftAND*	0	4	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+2.54	[+1.94, +3.13]
ml	SoftAND*	0	4	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+2.43	[+1.88, +2.97]
ml	SoftAND*	0	4	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+2.42	[+1.86, +2.98]
ml	SoftAND*	0	10	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+3.16	[+2.56, +3.76]
ml	SoftAND*	0	10	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+3.41	[+2.65, +4.17]
ml	SoftAND*	0	10	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.51	[+2.71, +4.32]
ml	SoftAND*	0	10	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+3.43	[+2.62, +4.23]
ml	SoftAND*	0	10	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+3.38	[+2.57, +4.20]
ml	SoftAND*	0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+3.92	[+3.22, +4.62]
ml	SoftAND*	0	100	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+3.49	[+2.67, +4.32]
ml	SoftAND*	0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.38	[+2.50, +4.25]
ml	SoftAND*	0	100	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+3.45	[+2.53, +4.38]
ml	SoftAND*	0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+3.40	[+2.48, +4.32]
ml	SoftAND*-adapt	0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+3.71	[+2.59, +4.83]
ml	SoftAND*-adapt	0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+3.80	[+2.33, +5.26]
ml	SoftAND*-adapt	0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+3.64	[+2.22, +5.05]
ml	SoftThresh-adapt	0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+6.20	[+5.11, +7.29]
ml	SoftThresh-adapt	0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+6.71	[+5.22, +8.21]
ml	SoftThresh-adapt	0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+6.39	[+5.22, +7.55]
ml	SoftThreshτ1.5	0	0.25	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-0.04	[-0.27, +0.19]
ml	SoftThreshτ1.5	0	0.25	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.28	[+0.01, +0.55]
ml	SoftThreshτ1.5	0	0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.08	[-0.20, +0.36]
ml	SoftThreshτ1.5	0	0.25	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.24	[+0.00, +0.48]
ml	SoftThreshτ1.5	0	0.25	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.33	[+0.08, +0.58]
ml	SoftThreshτ1.5	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-0.02	[-0.32, +0.28]
ml	SoftThreshτ1.5	0	0.5	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+0.82	[+0.41, +1.24]
ml	SoftThreshτ1.5	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.66	[+0.25, +1.07]
ml	SoftThreshτ1.5	0	0.5	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+0.86	[+0.46, +1.27]
ml	SoftThreshτ1.5	0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+0.87	[+0.49, +1.25]
ml	SoftThreshτ1.5	0	0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.13	[-0.29, +0.54]
ml	SoftThreshτ1.5	0	0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.30	[+0.87, +1.73]
ml	SoftThreshτ1.5	0	1	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.33	[-0.16, +0.81]
ml	SoftThreshτ1.5	0	1	$V_{f0.25} - V_{\mathrm{NLL}}$	34	+1.26	[+0.76, +1.77]
ml	SoftThreshτ1.5	0	1	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.75	[+1.21, +2.30]
ml	SoftThreshτ1.5	0	1	$V_{f0.75} - V_{\mathrm{NLL}}$	34	+1.71	[+1.19, +2.24]
ml	SoftThreshτ1.5	0	1	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	34	+1.68	[+1.12, +2.24]
ml	SoftThreshτ1.5	0	2	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.99	[+1.32, +2.65]
ml	SoftThreshτ1.5	0	2	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+3.24	[+2.49, +3.99]
ml	SoftThreshτ1.5	0	2	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.50	[+2.86, +4.15]
ml	SoftThreshτ1.5	0	2	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+3.04	[+2.47, +3.61]
ml	SoftThreshτ1.5	0	2	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+3.11	[+2.50, +3.71]
ml	SoftThreshτ1.5	0	4	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+4.31	[+3.60, +5.02]
ml	SoftThreshτ1.5	0	4	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+5.51	[+4.82, +6.20]
ml	SoftThreshτ1.5	0	10	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+5.75	[+4.91, +6.60]
ml	SoftThreshτ1.5	0	10	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+6.88	[+5.97, +7.79]
ml	SoftThreshτ1.5	0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+5.95	[+4.98, +6.91]
ml	SoftThreshτ1.5	0	100	$V_{f0.25} - V_{\mathrm{NLL}}$	34	+6.53	[+5.43, +7.62]
ml	SoftThreshτ1.5	0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+7.09	[+6.06, +8.12]
ml	SoftThreshτ1.5	0	100	$V_{f0.75} - V_{\mathrm{NLL}}$	34	+6.21	[+5.41, +7.02]
ml	SoftThreshτ1.5	0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	34	+6.13	[+5.29, +6.97]
myket	R*	0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.55	[-0.20, +1.31]

dataset	misspec	γ_0	γ_1	contrast	n	DiD	95% CI
myket	R*	0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.69	[+0.16, +1.22]

6 Results by setup

Each subsection is one (**dataset**, **misspec**) pair. Within it: absolute levels by γ and method, then paired contrasts at each γ, then DiD vs the family's γ=0 baseline (with 95% CI).

6.1 anime / R*

Gammas present: $0, 0.6$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	15.93	15.51	16.34
0	20	NLL	16.49	15.83	17.14
0	20	NLL→DR (ref)	12.13	10.54	13.73
0	20	hybrid f=0.5	16.71	15.93	17.48
0.6	20	OPC/DR	16.19	15.61	16.76
0.6	20	NLL	16.65	16.02	17.27
0.6	20	NLL→DR (ref)	12.39	10.64	14.14
0.6	20	hybrid f=0.5	17.37	16.60	18.13

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-0.56	-0.95	-0.17
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.22	-0.11	+0.55
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.78	+0.27	+1.29
0	H-nll_dr	20	+4.57	+2.99	+6.15
0.6	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-0.46	-0.99	+0.08
0.6	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.72	+0.39	+1.05
0.6	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+1.18	+0.52	+1.83
0.6	H-nll_dr	20	+4.98	+3.47	+6.48

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.6	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.10	-0.26	+0.46
0	0.6	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.50	+0.08	+0.92
0	0.6	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.40	-0.09	+0.89
0	0.6	H-nll_dr	20	+0.40	-0.16	+0.97

6.2 anime / SoftAND*

Gammas present: $0, 0.25, 0.75$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	33	OPC/DR	15.92	15.56	16.28
0	33	NLL	16.31	15.70	16.91
0	33	hybrid f=0.25	16.18	15.55	16.82
0	33	hybrid f=0.5	16.44	15.77	17.11
0	33	hybrid f=0.75	16.26	15.38	17.14
0	33	hybrid adapt	16.33	15.44	17.22
0.25	9	OPC/DR	16.23	15.05	17.41
0.25	9	NLL	16.84	15.91	17.76
0.25	9	hybrid f=0.25	16.63	15.05	18.22
0.25	9	hybrid f=0.5	17.19	15.51	18.87
0.25	9	hybrid f=0.75	17.09	14.84	19.35
0.25	9	hybrid adapt	17.06	14.82	19.29
0.75	5	OPC/DR	16.73	14.39	19.06
0.75	5	NLL	17.84	14.94	20.73
0.75	5	hybrid f=0.25	17.41	14.54	20.28
0.75	5	hybrid f=0.5	18.75	16.51	20.99
0.75	5	hybrid f=0.75	18.77	16.09	21.44
0.75	5	hybrid adapt	17.59	12.25	22.92

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	9	-0.24	-1.24	+0.76
0	$V_{f0.25} - V_{\mathrm{NLL}}$	9	-0.25	-1.59	+1.09
0	$V_{f0.5} - V_{\mathrm{NLL}}$	9	+0.08	-1.12	+1.28
0	$V_{f0.75} - V_{\mathrm{NLL}}$	9	+0.07	-1.58	+1.71
0	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	9	+0.05	-1.55	+1.65
0	$V_{f0.5} - V_{\mathrm{OPC}}$	9	+0.32	-0.47	+1.11
0	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	9	+0.29	-1.32	+1.89
0	$V_{\mathrm{adapt}} - V_{f0.5}$	9	-0.03	-1.16	+1.09
0.25	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	9	-0.60	-1.70	+0.49
0.25	$V_{f0.25} - V_{\mathrm{NLL}}$	9	-0.20	-1.72	+1.32
0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	9	+0.35	-1.04	+1.75
0.25	$V_{f0.75} - V_{\mathrm{NLL}}$	9	+0.26	-1.56	+2.08
0.25	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	9	+0.22	-1.59	+2.03
0.25	$V_{f0.5} - V_{\mathrm{OPC}}$	9	+0.96	+0.26	+1.65
0.25	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	9	+0.82	-0.78	+2.43
0.25	$V_{\mathrm{adapt}} - V_{f0.5}$	9	-0.13	-1.49	+1.22
0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	5	-1.11	-3.34	+1.13
0.75	$V_{f0.25} - V_{\mathrm{NLL}}$	5	-0.43	-2.29	+1.44
0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	5	+0.91	-0.78	+2.61
0.75	$V_{f0.75} - V_{\mathrm{NLL}}$	5	+0.93	-0.45	+2.31
0.75	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	5	-0.25	-4.17	+3.67
0.75	$V_{f0.5} - V_{\mathrm{OPC}}$	5	+2.02	+0.92	+3.13
0.75	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	5	+0.86	-3.82	+5.54
0.75	$V_{\mathrm{adapt}} - V_{f0.5}$	5	-1.17	-4.86	+2.53

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.25	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	9	-0.36	-0.71	-0.02
0	0.25	$V_{f0.25} - V_{\mathrm{NLL}}$	9	+0.05	-0.38	+0.47

continued: anime / SoftAND DiD*

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	9	+0.27	-0.13	+0.68
0	0.25	$V_{f0.75} - V_{\mathrm{NLL}}$	9	+0.19	-0.35	+0.72
0	0.25	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	9	+0.17	-0.43	+0.78
0	0.25	$V_{f0.5} - V_{\mathrm{OPC}}$	9	+0.63	+0.16	+1.11
0	0.25	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	9	+0.54	-0.03	+1.11
0	0.25	$V_{\mathrm{adapt}} - V_{f0.5}$	9	-0.10	-0.59	+0.40
0	0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	5	-0.50	-1.53	+0.53
0	0.75	$V_{f0.25} - V_{\mathrm{NLL}}$	5	+0.18	-1.20	+1.55
0	0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	5	+1.47	-0.33	+3.27
0	0.75	$V_{f0.75} - V_{\mathrm{NLL}}$	5	+1.79	-0.53	+4.10
0	0.75	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	5	+0.60	-1.62	+2.82
0	0.75	$V_{f0.5} - V_{\mathrm{OPC}}$	5	+1.97	+1.20	+2.74
0	0.75	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	5	+1.10	-0.54	+2.75
0	0.75	$V_{\mathrm{adapt}} - V_{f0.5}$	5	-0.87	-2.43	+0.69

6.3 kuairand / R*-full

Gammas present: $0, 0.5$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	20.78	20.36	21.19
0	20	NLL	22.15	21.78	22.52
0	20	NLL→DR (ref)	22.15	21.78	22.53
0	20	hybrid f=0.5	21.57	21.29	21.85
0.5	20	OPC/DR	20.62	19.90	21.35
0.5	20	NLL	21.83	21.04	22.62
0.5	20	NLL→DR (ref)	21.82	21.03	22.61
0.5	20	hybrid f=0.5	21.86	21.06	22.66

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-1.38	-1.75	-1.00
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	-0.58	-0.87	-0.30
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.79	+0.38	+1.21
0	H-nll_dr	20	-0.58	-0.87	-0.30
0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-1.21	-1.66	-0.76
0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.03	-0.41	+0.47
0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+1.24	+0.83	+1.65
0.5	H-nll_dr	20	+0.05	-0.39	+0.49

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.17	-0.28	+0.62
0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.61	+0.12	+1.11
0	0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.45	-0.04	+0.93
0	0.5	H-nll_dr	20	+0.63	+0.13	+1.13

6.4 kuairand / R*-mlsize

Gammas present: $0, 0.5$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	20.00	19.52	20.49
0	20	NLL	21.57	21.15	21.98
0	20	NLL→DR (ref)	21.58	21.17	21.98
0	20	hybrid f=0.5	20.79	20.34	21.23
0.5	20	OPC/DR	20.95	20.33	21.57
0.5	20	NLL	21.34	20.69	21.98
0.5	20	NLL→DR (ref)	21.41	20.78	22.05
0.5	20	hybrid f=0.5	21.86	21.21	22.51

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-1.57	-1.97	-1.16
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	-0.78	-1.04	-0.52
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.79	+0.39	+1.19
0	H-nll_dr	20	-0.79	-1.05	-0.52
0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-0.39	-0.79	+0.02
0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.53	+0.14	+0.92
0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.92	+0.51	+1.32
0.5	H-nll_dr	20	+0.45	+0.07	+0.83

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+1.18	+0.69	+1.67
0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+1.31	+0.92	+1.69
0	0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.13	-0.30	+0.56
0	0.5	H-nll_dr	20	+1.24	+0.83	+1.65

6.5 kuairec / R*-full

Gammas present: $0, 0.5$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	20.54	19.74	21.34
0	20	NLL	22.27	21.55	23.00
0	20	NLL→DR (ref)	22.24	21.48	22.99
0	20	hybrid f=0.5	21.59	20.67	22.50
0.5	20	OPC/DR	21.81	20.96	22.65
0.5	20	NLL	21.98	20.90	23.07
0.5	20	NLL→DR (ref)	21.92	20.83	23.02
0.5	20	hybrid f=0.5	22.60	21.53	23.67

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-1.74	-2.22	-1.25
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	-0.68	-1.24	-0.12
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+1.05	+0.48	+1.62
0	H-nll_dr	20	-0.65	-1.22	-0.07
0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-0.18	-0.74	+0.39
0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.62	+0.09	+1.14
0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.79	+0.34	+1.24
0.5	H-nll_dr	20	+0.68	+0.13	+1.22

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+1.56	+0.92	+2.20
0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+1.30	+0.72	+1.88
0	0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	-0.26	-0.88	+0.36
0	0.5	H-nll_dr	20	+1.32	+0.74	+1.91

6.6 kuairec / R*-mlsize

Gammas present: $0, 0.5$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	22.02	21.40	22.63
0	20	NLL	23.01	22.25	23.76
0	20	NLL→DR (ref)	23.00	22.25	23.75
0	20	hybrid f=0.5	22.87	22.22	23.53
0.5	20	OPC/DR	22.24	21.41	23.08
0.5	20	NLL	22.29	21.34	23.24
0.5	20	NLL→DR (ref)	22.27	21.31	23.23
0.5	20	hybrid f=0.5	22.87	21.86	23.88

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-0.99	-1.56	-0.42
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	-0.13	-0.80	+0.53
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.85	+0.17	+1.54
0	H-nll_dr	20	-0.13	-0.79	+0.54
0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-0.05	-0.47	+0.38
0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.58	+0.12	+1.05
0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.63	+0.10	+1.16
0.5	H-nll_dr	20	+0.60	+0.13	+1.07

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.94	+0.46	+1.42
0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.72	-0.06	+1.50
0	0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	-0.22	-1.00	+0.56
0	0.5	H-nll_dr	20	+0.73	-0.05	+1.51

6.7 ml / R*

Gammas present: $0, 0.25, 0.5, 0.75, 1, 2, 4, 10, 100$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	40	OPC/DR	17.35	17.06	17.63
0	40	NLL	18.54	18.16	18.92
0	20	NLL→DR (ref)	15.54	13.38	17.70
0	36	hybrid f=0.25	18.03	17.63	18.43
0	40	hybrid f=0.5	18.37	18.08	18.65
0	36	hybrid f=0.75	18.42	18.08	18.76
0	36	hybrid adapt	18.41	18.05	18.78
0.25	40	OPC/DR	18.01	17.59	18.42
0.25	40	NLL	18.94	18.41	19.46
0.25	40	hybrid f=0.5	19.13	18.64	19.61
0.5	40	OPC/DR	19.05	18.58	19.52
0.5	40	NLL	19.12	18.49	19.74
0.5	20	NLL→DR (ref)	16.20	13.92	18.48
0.5	10	hybrid f=0.25	19.10	17.81	20.40
0.5	40	hybrid f=0.5	20.06	19.43	20.68
0.5	10	hybrid f=0.75	19.56	18.76	20.36
0.75	40	OPC/DR	19.65	19.14	20.16
0.75	40	NLL	18.91	18.21	19.61
0.75	40	hybrid f=0.25	20.15	19.46	20.84
0.75	40	hybrid f=0.5	20.47	19.78	21.16
0.75	40	hybrid f=0.75	20.45	19.76	21.14
0.75	40	hybrid adapt	20.53	19.82	21.23
1	40	OPC/DR	20.04	19.46	20.62
1	40	NLL	18.64	17.89	19.38
1	40	hybrid f=0.25	20.54	19.78	21.30
1	40	hybrid f=0.5	20.71	20.00	21.41
1	40	hybrid f=0.75	20.71	19.94	21.48
1	40	hybrid adapt	20.80	20.02	21.57
2	40	OPC/DR	20.35	19.74	20.96
2	40	NLL	17.08	16.37	17.78
2	40	hybrid f=0.25	19.97	19.10	20.84
2	40	hybrid f=0.5	20.48	19.61	21.36
2	40	hybrid f=0.75	20.25	19.36	21.13
2	40	hybrid adapt	20.20	19.35	21.04
4	40	OPC/DR	20.02	19.44	20.60
4	40	NLL	15.65	15.01	16.29
4	40	hybrid f=0.25	19.44	18.53	20.35
4	40	hybrid f=0.5	19.64	18.78	20.50
4	40	hybrid f=0.75	19.65	18.78	20.51
4	40	hybrid adapt	19.58	18.65	20.50
10	40	OPC/DR	19.59	18.96	20.22
10	40	NLL	14.70	14.06	15.33
10	40	hybrid f=0.25	18.86	17.95	19.77
10	40	hybrid f=0.5	19.27	18.30	20.24
10	40	hybrid f=0.75	18.97	18.04	19.90
10	40	hybrid adapt	18.89	17.98	19.79
100	40	OPC/DR	19.39	18.75	20.03
100	40	NLL	14.13	13.48	14.78
100	40	hybrid f=0.25	18.40	17.51	19.30
100	40	hybrid f=0.5	18.68	17.86	19.51
100	40	hybrid f=0.75	18.42	17.53	19.31
100	40	hybrid adapt	18.33	17.38	19.27

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-1.19	-1.53	-0.85
0	$V_{f0.25} - V_{\mathrm{NLL}}$	36	-0.53	-0.85	-0.20
0	$V_{f0.5} - V_{\mathrm{NLL}}$	40	-0.17	-0.40	+0.06
0	$V_{f0.75} - V_{\mathrm{NLL}}$	36	-0.13	-0.41	+0.14
0	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	-0.14	-0.42	+0.14
0	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.02	+0.73	+1.31
0	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	36	+1.11	+0.72	+1.50
0	$V_{\mathrm{adapt}} - V_{f0.5}$	36	+0.04	-0.15	+0.22
0	H-nll_dr	20	+2.53	+0.52	+4.54
0.25	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-0.93	-1.31	-0.55
0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.19	-0.11	+0.50
0.25	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.12	+0.83	+1.41
0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-0.07	-0.50	+0.37
0.5	$V_{f0.25} - V_{\mathrm{NLL}}$	10	+1.14	-0.14	+2.42
0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.94	+0.53	+1.35
0.5	$V_{f0.75} - V_{\mathrm{NLL}}$	10	+1.60	+0.83	+2.36
0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.01	+0.56	+1.46
0.5	H-nll_dr	20	+3.42	+1.30	+5.54
0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.82	+0.38	+1.25
0.75	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+1.23	+0.71	+1.74
0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.56	+1.07	+2.06
0.75	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+1.54	+1.22	+1.85
0.75	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+1.62	+1.29	+1.94
0.75	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.75	+0.27	+1.22
0.75	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	36	+0.88	+0.35	+1.40
0.75	$V_{\mathrm{adapt}} - V_{f0.5}$	36	+0.06	-0.32	+0.43
1	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.47	+1.05	+1.89
1	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+1.93	+1.42	+2.44
1	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+2.11	+1.62	+2.59
1	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+2.03	+1.63	+2.44
1	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+2.11	+1.65	+2.57
1	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.64	+0.22	+1.06
1	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	36	+0.69	+0.19	+1.20
1	$V_{\mathrm{adapt}} - V_{f0.5}$	36	+0.08	-0.16	+0.31
2	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+3.26	+2.86	+3.67
2	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+2.95	+2.25	+3.65
2	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.46	+2.87	+4.04
2	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+3.20	+2.64	+3.75
2	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+3.14	+2.62	+3.66
2	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.19	-0.32	+0.70
2	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	36	-0.17	-0.73	+0.40
2	$V_{\mathrm{adapt}} - V_{f0.5}$	36	-0.29	-0.67	+0.09
4	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+4.37	+3.96	+4.78
4	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+3.80	+3.04	+4.55
4	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+4.05	+3.38	+4.71
4	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+4.06	+3.38	+4.74
4	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+4.01	+3.31	+4.72
4	$V_{f0.5} - V_{\mathrm{OPC}}$	40	-0.32	-0.92	+0.28
4	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	36	-0.39	-0.98	+0.19
4	$V_{\mathrm{adapt}} - V_{f0.5}$	36	-0.02	-0.50	+0.46
10	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	36	+4.99	+4.55	+5.43
10	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+4.20	+3.45	+4.95
10	$V_{f0.5} - V_{\mathrm{NLL}}$	36	+4.66	+3.86	+5.46
10	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+4.36	+3.60	+5.12
10	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+4.24	+3.54	+4.95
10	$V_{f0.5} - V_{\mathrm{OPC}}$	36	-0.33	-0.98	+0.31
10	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	36	-0.75	-1.32	-0.18
10	$V_{\mathrm{adapt}} - V_{f0.5}$	36	-0.41	-0.89	+0.06
100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	36	+5.30	+4.76	+5.85
100	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+4.29	+3.54	+5.04
100	$V_{f0.5} - V_{\mathrm{NLL}}$	36	+4.58	+3.89	+5.26
100	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+4.31	+3.63	+4.99
100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+4.22	+3.44	+4.99
100	$V_{f0.5} - V_{\mathrm{OPC}}$	36	-0.73	-1.25	-0.20
100	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	36	-1.09	-1.67	-0.50

continued: ml / R^* contrasts

γ	contrast	n	mean	95% CI lo	95% CI hi
100	$V_{\text{adapt}} - V_{f0.5}$	36	-0.36	-0.81	+0.09

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.25	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+0.26	+0.03	+0.49
0	0.25	$V_{f0.5} - V_{\text{NLL}}$	40	+0.36	+0.12	+0.61
0	0.25	$V_{f0.5} - V_{\text{OPC}}$	40	+0.10	-0.15	+0.35
0	0.5	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+1.12	+0.78	+1.47
0	0.5	$V_{f0.25} - V_{\text{NLL}}$	10	+1.82	+0.58	+3.05
0	0.5	$V_{f0.5} - V_{\text{NLL}}$	40	+1.11	+0.72	+1.50
0	0.5	$V_{f0.75} - V_{\text{NLL}}$	10	+1.71	+1.02	+2.39
0	0.5	$V_{f0.5} - V_{\text{OPC}}$	40	-0.01	-0.45	+0.42
0	0.5	H-nll_dr	20	+0.90	+0.10	+1.69
0	0.75	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+2.01	+1.59	+2.42
0	0.75	$V_{f0.25} - V_{\text{NLL}}$	36	+1.75	+1.20	+2.31
0	0.75	$V_{f0.5} - V_{\text{NLL}}$	40	+1.73	+1.23	+2.24
0	0.75	$V_{f0.75} - V_{\text{NLL}}$	36	+1.67	+1.28	+2.05
0	0.75	$V_{\text{adapt}} - V_{\text{NLL}}$	36	+1.76	+1.37	+2.15
0	0.75	$V_{f0.5} - V_{\text{OPC}}$	40	-0.27	-0.70	+0.15
0	0.75	$V_{\text{adapt}} - V_{\text{OPC}}$	36	-0.23	-0.75	+0.29
0	0.75	$V_{\text{adapt}} - V_{f0.5}$	36	+0.02	-0.41	+0.45
0	1	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+2.66	+2.22	+3.09
0	1	$V_{f0.25} - V_{\text{NLL}}$	36	+2.45	+1.90	+3.01
0	1	$V_{f0.5} - V_{\text{NLL}}$	40	+2.28	+1.78	+2.77
0	1	$V_{f0.75} - V_{\text{NLL}}$	36	+2.17	+1.69	+2.64
0	1	$V_{\text{adapt}} - V_{\text{NLL}}$	36	+2.25	+1.74	+2.76
0	1	$V_{f0.5} - V_{\text{OPC}}$	40	-0.38	-0.84	+0.08
0	1	$V_{\text{adapt}} - V_{\text{OPC}}$	36	-0.42	-0.94	+0.10
0	1	$V_{\text{adapt}} - V_{f0.5}$	36	+0.04	-0.23	+0.31
0	2	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+4.46	+3.96	+4.96
0	2	$V_{f0.25} - V_{\text{NLL}}$	36	+3.48	+2.69	+4.27
0	2	$V_{f0.5} - V_{\text{NLL}}$	40	+3.63	+2.95	+4.31
0	2	$V_{f0.75} - V_{\text{NLL}}$	36	+3.33	+2.68	+3.98
0	2	$V_{\text{adapt}} - V_{\text{NLL}}$	36	+3.28	+2.69	+3.88
0	2	$V_{f0.5} - V_{\text{OPC}}$	40	-0.83	-1.34	-0.32
0	2	$V_{\text{adapt}} - V_{\text{OPC}}$	36	-1.28	-1.89	-0.66
0	2	$V_{\text{adapt}} - V_{f0.5}$	36	-0.33	-0.78	+0.12
0	4	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+5.56	+5.02	+6.09
0	4	$V_{f0.25} - V_{\text{NLL}}$	36	+4.32	+3.46	+5.18
0	4	$V_{f0.5} - V_{\text{NLL}}$	40	+4.22	+3.46	+4.98
0	4	$V_{f0.75} - V_{\text{NLL}}$	36	+4.19	+3.44	+4.94
0	4	$V_{\text{adapt}} - V_{\text{NLL}}$	36	+4.15	+3.39	+4.91
0	4	$V_{f0.5} - V_{\text{OPC}}$	40	-1.34	-1.98	-0.69
0	4	$V_{\text{adapt}} - V_{\text{OPC}}$	36	-1.50	-2.16	-0.85
0	4	$V_{\text{adapt}} - V_{f0.5}$	36	-0.06	-0.58	+0.47
0	10	$V_{\text{OPC}} - V_{\text{NLL}}$	36	+6.24	+5.67	+6.82
0	10	$V_{f0.25} - V_{\text{NLL}}$	36	+4.72	+3.88	+5.57
0	10	$V_{f0.5} - V_{\text{NLL}}$	36	+4.84	+3.93	+5.74
0	10	$V_{f0.75} - V_{\text{NLL}}$	36	+4.49	+3.66	+5.32
0	10	$V_{\text{adapt}} - V_{\text{NLL}}$	36	+4.39	+3.62	+5.15
0	10	$V_{f0.5} - V_{\text{OPC}}$	36	-1.40	-2.10	-0.71
0	10	$V_{\text{adapt}} - V_{\text{OPC}}$	36	-1.86	-2.46	-1.25
0	10	$V_{\text{adapt}} - V_{f0.5}$	36	-0.45	-0.96	+0.06
0	100	$V_{\text{OPC}} - V_{\text{NLL}}$	36	+6.55	+5.89	+7.21
0	100	$V_{f0.25} - V_{\text{NLL}}$	36	+4.81	+3.95	+5.68
0	100	$V_{f0.5} - V_{\text{NLL}}$	36	+4.76	+3.95	+5.56
0	100	$V_{f0.75} - V_{\text{NLL}}$	36	+4.44	+3.66	+5.23
0	100	$V_{\text{adapt}} - V_{\text{NLL}}$	36	+4.36	+3.51	+5.21
0	100	$V_{f0.5} - V_{\text{OPC}}$	36	-1.80	-2.38	-1.22

continued: ml / R DiD*

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	100	$V_{\text{adapt}} - V_{\text{OPC}}$	36	-2.20	-2.86	-1.53
0	100	$V_{\text{adapt}} - V_{f0.5}$	36	-0.40	-0.87	+0.07

6.8 ml / R*-adapt

Gammas present: $0, 0.5$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	17.17	16.79	17.54
0	20	NLL	18.27	17.74	18.81
0	20	hybrid f=0.5	18.07	17.71	18.42
0	20	hybrid adapt	18.05	17.57	18.54
0.5	20	OPC/DR	18.88	18.32	19.44
0.5	20	NLL	18.63	17.86	19.41
0.5	20	hybrid f=0.5	19.62	18.98	20.27
0.5	20	hybrid adapt	19.86	19.18	20.53

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-1.11	-1.62	-0.59
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	-0.20	-0.51	+0.10
0	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	-0.22	-0.60	+0.16
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.90	+0.52	+1.28
0	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	+0.89	+0.33	+1.45
0	$V_{\mathrm{adapt}} - V_{f0.5}$	20	-0.01	-0.27	+0.24
0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.25	-0.29	+0.79
0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.99	+0.33	+1.65
0.5	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+1.22	+0.83	+1.61
0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.74	+0.17	+1.31
0.5	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	+0.98	+0.46	+1.49
0.5	$V_{\mathrm{adapt}} - V_{f0.5}$	20	+0.24	-0.08	+0.56

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD $= \Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+1.35	+0.88	+1.83
0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+1.19	+0.59	+1.79
0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+1.44	+1.03	+1.85
0	0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	-0.16	-0.71	+0.38
0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	+0.09	-0.45	+0.63
0	0.5	$V_{\mathrm{adapt}} - V_{f0.5}$	20	+0.25	-0.16	+0.67

6.9 ml / SoftAND*

Gammas present: $0, 0.25, 0.5, 0.75, 1, 2, 4, 10, 100$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	40	OPC/DR	17.32	17.03	17.60
0	40	NLL	18.54	18.16	18.92
0	10	NLL→DR (ref)	16.99	14.26	19.71
0	40	hybrid f=0.25	18.00	17.63	18.37
0	40	hybrid f=0.5	18.36	18.07	18.64
0	40	hybrid f=0.75	18.42	18.08	18.76
0	40	hybrid adapt	18.40	18.04	18.76
0.25	40	OPC/DR	17.49	17.17	17.81
0.25	40	NLL	18.71	18.33	19.09
0.25	40	hybrid f=0.25	18.43	18.04	18.82
0.25	40	hybrid f=0.5	18.61	18.25	18.98
0.25	40	hybrid f=0.75	18.78	18.42	19.14
0.25	40	hybrid adapt	18.79	18.44	19.15
0.5	40	OPC/DR	17.54	17.24	17.84
0.5	40	NLL	18.64	18.23	19.04
0.5	10	NLL→DR (ref)	17.13	14.28	19.98
0.5	40	hybrid f=0.25	18.49	18.04	18.94
0.5	40	hybrid f=0.5	18.83	18.41	19.24
0.5	40	hybrid f=0.75	18.93	18.52	19.35
0.5	40	hybrid adapt	18.87	18.46	19.29
0.75	40	OPC/DR	17.64	17.30	17.97
0.75	40	NLL	18.51	18.08	18.93
0.75	40	hybrid f=0.25	18.68	18.23	19.13
0.75	40	hybrid f=0.5	18.87	18.46	19.28
0.75	40	hybrid f=0.75	19.09	18.60	19.58
0.75	40	hybrid adapt	19.12	18.62	19.62
1	40	OPC/DR	17.68	17.32	18.04
1	40	NLL	18.43	18.00	18.87
1	40	hybrid f=0.25	18.63	18.16	19.09
1	40	hybrid f=0.5	19.07	18.62	19.53
1	40	hybrid f=0.75	19.13	18.62	19.64
1	40	hybrid adapt	19.10	18.57	19.63
2	40	OPC/DR	18.25	17.79	18.72
2	40	NLL	18.42	17.92	18.93
2	40	hybrid f=0.25	19.36	18.70	20.03
2	40	hybrid f=0.5	19.95	19.34	20.57
2	40	hybrid f=0.75	19.83	19.26	20.39
2	40	hybrid adapt	19.81	19.22	20.41
4	40	OPC/DR	19.32	18.81	19.82
4	40	NLL	18.66	18.03	19.29
4	40	hybrid f=0.25	20.55	19.76	21.35
4	40	hybrid f=0.5	21.02	20.26	21.77
4	40	hybrid f=0.75	20.97	20.26	21.67
4	40	hybrid adapt	20.94	20.24	21.64
10	40	OPC/DR	20.62	20.03	21.21
10	40	NLL	18.68	17.96	19.40
10	40	hybrid f=0.25	21.55	20.67	22.44
10	40	hybrid f=0.5	22.02	21.15	22.88
10	40	hybrid f=0.75	21.99	21.09	22.89
10	40	hybrid adapt	21.93	21.02	22.83
100	40	OPC/DR	20.85	20.27	21.43
100	40	NLL	18.11	17.43	18.80
100	10	NLL→DR (ref)	18.01	16.60	19.42
100	40	hybrid f=0.25	21.07	20.15	21.99
100	40	hybrid f=0.5	21.40	20.54	22.27
100	40	hybrid f=0.75	21.45	20.55	22.34
100	40	hybrid adapt	21.37	20.48	22.27

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-1.19	-1.53	-0.85
0	$V_{f0.25} - V_{\mathrm{NLL}}$	40	-0.54	-0.83	-0.25
0	$V_{f0.5} - V_{\mathrm{NLL}}$	40	-0.17	-0.40	+0.06
0	$V_{f0.75} - V_{\mathrm{NLL}}$	40	-0.12	-0.37	+0.13
0	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	-0.14	-0.40	+0.12
0	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.02	+0.73	+1.31
0	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.08	+0.70	+1.46
0	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.04	-0.13	+0.21
0	H-nll_dr	10	+1.24	-1.28	+3.76
0.25	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-1.22	-1.59	-0.84
0.25	$V_{f0.25} - V_{\mathrm{NLL}}$	40	-0.28	-0.58	+0.02
0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	40	-0.10	-0.35	+0.16
0.25	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.07	-0.13	+0.27
0.25	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.08	-0.12	+0.28
0.25	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.12	+0.82	+1.43
0.25	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.30	+0.98	+1.62
0.25	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.18	-0.04	+0.39
0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-1.06	-1.39	-0.74
0.5	$V_{f0.25} - V_{\mathrm{NLL}}$	40	-0.15	-0.42	+0.12
0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.18	-0.03	+0.40
0.5	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.30	+0.11	+0.48
0.5	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.24	+0.06	+0.41
0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.25	+0.95	+1.54
0.5	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.34	+1.01	+1.66
0.5	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.05	-0.13	+0.23
0.5	H-nll_dr	10	+1.84	-0.45	+4.13
0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-0.87	-1.22	-0.52
0.75	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.18	-0.14	+0.50
0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.36	+0.08	+0.65
0.75	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.58	+0.38	+0.79
0.75	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.62	+0.40	+0.83
0.75	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.23	+0.86	+1.60
0.75	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.48	+1.08	+1.89
0.75	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.25	-0.01	+0.52
1	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-0.75	-1.11	-0.39
1	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.19	-0.12	+0.50
1	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.64	+0.36	+0.92
1	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.70	+0.45	+0.95
1	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.67	+0.41	+0.93
1	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.39	+0.98	+1.81
1	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.42	+0.96	+1.88
1	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.03	-0.26	+0.31
2	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-0.17	-0.57	+0.23
2	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.94	+0.48	+1.39
2	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.53	+1.16	+1.90
2	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+1.40	+1.15	+1.66
2	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+1.39	+1.10	+1.68
2	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.70	+1.16	+2.23
2	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.56	+1.05	+2.07
2	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.14	-0.49	+0.21
4	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.66	+0.21	+1.10
4	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+1.89	+1.35	+2.43
4	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+2.36	+1.87	+2.84
4	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+2.31	+1.85	+2.76
4	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+2.28	+1.80	+2.75
4	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.70	+1.15	+2.25
4	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.62	+1.10	+2.15
4	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.08	-0.37	+0.21
10	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.94	+1.37	+2.51
10	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+2.87	+2.19	+3.56
10	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.33	+2.57	+4.10
10	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+3.31	+2.52	+4.09

continued: ml / SoftAND contrasts*					
γ	contrast	n	mean	95% CI lo	95% CI hi
10	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+3.24	+2.46	+4.02
10	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.39	+0.80	+1.99
10	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.30	+0.71	+1.90
10	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.09	-0.47	+0.29
100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+2.73	+2.05	+3.41
100	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+2.95	+2.15	+3.76
100	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.20	+2.36	+4.05
100	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+3.33	+2.44	+4.22
100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+3.26	+2.38	+4.14
100	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.48	-0.15	+1.10
100	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.52	-0.14	+1.19
100	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.03	-0.42	+0.36
100	H-nll_dr	10	+3.78	+1.30	+6.26

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.25	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.00	-0.19	+0.19
0	0.25	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.26	+0.06	+0.46
0	0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.09	-0.11	+0.28
0	0.25	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.19	-0.03	+0.41
0	0.25	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.22	-0.00	+0.45
0	0.25	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.09	-0.15	+0.32
0	0.25	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.22	-0.07	+0.51
0	0.25	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.14	-0.09	+0.36
0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.13	-0.08	+0.34
0	0.5	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.39	+0.15	+0.63
0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.35	+0.13	+0.58
0	0.5	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.42	+0.15	+0.68
0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.38	+0.11	+0.65
0	0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.23	-0.03	+0.49
0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.26	-0.05	+0.56
0	0.5	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.01	-0.21	+0.23
0	0.5	H-nll_dr	10	+0.60	-0.00	+1.21
0	0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.35	+0.10	+0.60
0	0.75	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.71	+0.41	+1.02
0	0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.55	+0.23	+0.86
0	0.75	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.70	+0.38	+1.03
0	0.75	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.76	+0.43	+1.09
0	0.75	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.19	-0.15	+0.53
0	0.75	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.40	+0.02	+0.79
0	0.75	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.21	-0.09	+0.51
0	1	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.47	+0.21	+0.73
0	1	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.73	+0.34	+1.12
0	1	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.82	+0.52	+1.13
0	1	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.82	+0.46	+1.18
0	1	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.81	+0.47	+1.15
0	1	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.35	+0.01	+0.69
0	1	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.34	-0.06	+0.74
0	1	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.02	-0.29	+0.26
0	2	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.05	+0.72	+1.38
0	2	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+1.47	+0.88	+2.06
0	2	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.71	+1.24	+2.18
0	2	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+1.52	+1.15	+1.90
0	2	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+1.53	+1.10	+1.96
0	2	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.66	+0.22	+1.10
0	2	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.48	+0.01	+0.95
0	2	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.18	-0.57	+0.21
0	4	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.88	+1.43	+2.32
0	4	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+2.43	+1.77	+3.09

continued: ml / SoftAND DiD*

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	4	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+2.54	+1.94	+3.13
0	4	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+2.43	+1.88	+2.97
0	4	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+2.42	+1.86	+2.98
0	4	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.66	+0.10	+1.22
0	4	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.54	-0.01	+1.10
0	4	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.12	-0.44	+0.20
0	10	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+3.16	+2.56	+3.76
0	10	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+3.41	+2.65	+4.17
0	10	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.51	+2.71	+4.32
0	10	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+3.43	+2.62	+4.23
0	10	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+3.38	+2.57	+4.20
0	10	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.35	-0.27	+0.98
0	10	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.22	-0.43	+0.88
0	10	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.13	-0.53	+0.27
0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+3.92	+3.22	+4.62
0	100	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+3.49	+2.67	+4.32
0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.38	+2.50	+4.25
0	100	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+3.45	+2.53	+4.38
0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+3.40	+2.48	+4.32
0	100	$V_{f0.5} - V_{\mathrm{OPC}}$	40	-0.54	-1.21	+0.12
0	100	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	-0.56	-1.28	+0.17
0	100	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.07	-0.44	+0.30
0	100	H-nll_dr	10	+2.54	-1.58	+6.66

6.10 ml / SoftAND*-adapt

Gammas present: $0, 100$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	17.17	16.79	17.54
0	20	NLL	18.27	17.74	18.81
0	20	hybrid f=0.5	18.07	17.71	18.42
0	20	hybrid adapt	18.05	17.57	18.54
100	20	OPC/DR	21.09	20.21	21.98
100	20	NLL	18.49	17.64	19.34
100	20	hybrid f=0.5	22.08	20.85	23.31
100	20	hybrid adapt	21.91	20.58	23.24

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-1.11	-1.62	-0.59
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	-0.20	-0.51	+0.10
0	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	-0.22	-0.60	+0.16
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.90	+0.52	+1.28
0	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	+0.89	+0.33	+1.45
0	$V_{\mathrm{adapt}} - V_{f0.5}$	20	-0.01	-0.27	+0.24
100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+2.60	+1.62	+3.58
100	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+3.59	+2.22	+4.97
100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+3.42	+2.08	+4.76
100	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.99	-0.01	+1.99
100	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	+0.82	-0.16	+1.80
100	$V_{\mathrm{adapt}} - V_{f0.5}$	20	-0.17	-0.76	+0.42

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD $= \Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+3.71	+2.59	+4.83
0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+3.80	+2.33	+5.26
0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+3.64	+2.22	+5.05
0	100	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.09	-0.94	+1.12
0	100	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	-0.07	-1.16	+1.02
0	100	$V_{\mathrm{adapt}} - V_{f0.5}$	20	-0.16	-0.73	+0.41

6.11 ml / SoftThresh-adapt

Gammas present: $0, 100$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	17.17	16.79	17.54
0	20	NLL	18.27	17.74	18.81
0	20	hybrid f=0.5	18.07	17.71	18.42
0	20	hybrid adapt	18.05	17.57	18.54
100	20	OPC/DR	25.46	23.68	27.23
100	20	NLL	20.36	18.72	22.01
100	20	hybrid f=0.5	26.87	24.49	29.26
100	20	hybrid adapt	26.53	24.51	28.55

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-1.11	-1.62	-0.59
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	-0.20	-0.51	+0.10
0	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	-0.22	-0.60	+0.16
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.90	+0.52	+1.28
0	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	+0.89	+0.33	+1.45
0	$V_{\mathrm{adapt}} - V_{f0.5}$	20	-0.01	-0.27	+0.24
100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+5.09	+3.97	+6.22
100	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+6.51	+4.95	+8.07
100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+6.17	+4.96	+7.37
100	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+1.42	+0.26	+2.57
100	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	+1.07	+0.08	+2.06
100	$V_{\mathrm{adapt}} - V_{f0.5}$	20	-0.34	-1.16	+0.48

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD $= \Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+6.20	+5.11	+7.29
0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+6.71	+5.22	+8.21
0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	20	+6.39	+5.22	+7.55
0	100	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.51	-0.70	+1.73
0	100	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	20	+0.19	-0.76	+1.13
0	100	$V_{\mathrm{adapt}} - V_{f0.5}$	20	-0.33	-1.13	+0.47

6.12 ml / SoftThreshτ1.5

Gammas present: $0, 0.25, 0.5, 0.75, 1, 2, 4, 10, 100$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	40	OPC/DR	17.32	17.03	17.60
0	40	NLL	18.54	18.16	18.92
0	10	NLL→DR (ref)	16.99	14.26	19.71
0	40	hybrid f=0.25	18.00	17.63	18.37
0	40	hybrid f=0.5	18.36	18.07	18.64
0	40	hybrid f=0.75	18.42	18.08	18.76
0	40	hybrid adapt	18.40	18.04	18.76
0.25	40	OPC/DR	18.69	18.12	19.26
0.25	40	NLL	19.95	19.29	20.61
0.25	40	hybrid f=0.25	19.69	18.95	20.43
0.25	40	hybrid f=0.5	19.85	19.20	20.50
0.25	40	hybrid f=0.75	20.07	19.44	20.71
0.25	40	hybrid adapt	20.14	19.48	20.80
0.5	40	OPC/DR	19.93	19.07	20.80
0.5	40	NLL	21.17	20.31	22.03
0.5	36	hybrid f=0.25	21.50	20.50	22.49
0.5	40	hybrid f=0.5	21.65	20.66	22.64
0.5	36	hybrid f=0.75	21.93	20.97	22.88
0.5	36	hybrid adapt	21.93	21.00	22.85
0.75	40	OPC/DR	20.81	19.74	21.87
0.75	40	NLL	21.90	20.90	22.90
0.75	40	hybrid f=0.5	23.02	21.85	24.18
1	40	OPC/DR	21.41	20.29	22.53
1	40	NLL	22.30	21.26	23.34
1	34	hybrid f=0.25	22.95	21.76	24.14
1	40	hybrid f=0.5	23.87	22.58	25.17
1	34	hybrid f=0.75	23.76	22.62	24.90
1	34	hybrid adapt	23.72	22.57	24.87
2	40	OPC/DR	23.48	22.16	24.81
2	40	NLL	22.72	21.56	23.88
2	40	hybrid f=0.25	25.42	23.89	26.95
2	40	hybrid f=0.5	26.04	24.56	27.51
2	40	hybrid f=0.75	25.64	24.23	27.05
2	40	hybrid adapt	25.68	24.25	27.12
4	40	OPC/DR	25.10	23.77	26.43
4	40	NLL	22.01	20.77	23.26
4	40	hybrid f=0.5	27.34	25.91	28.77
10	40	OPC/DR	25.74	24.46	27.03
10	40	NLL	21.21	20.03	22.40
10	40	hybrid f=0.5	27.91	26.44	29.38
100	40	OPC/DR	25.36	24.12	26.60
100	40	NLL	20.63	19.51	21.75
100	10	NLL→DR (ref)	19.49	16.60	22.38
100	34	hybrid f=0.25	26.77	25.09	28.46
100	40	hybrid f=0.5	27.53	26.02	29.05
100	34	hybrid f=0.75	26.82	25.52	28.12
100	34	hybrid adapt	26.73	25.40	28.05

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-1.22	-1.56	-0.89
0	$V_{f0.25} - V_{\mathrm{NLL}}$	40	-0.54	-0.83	-0.25
0	$V_{f0.5} - V_{\mathrm{NLL}}$	40	-0.18	-0.40	+0.04
0	$V_{f0.75} - V_{\mathrm{NLL}}$	40	-0.12	-0.37	+0.13

continued: ml / SoftThreshτ1.5 contrasts

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\text{adapt}} - V_{\text{NLL}}$	40	-0.14	-0.40	+0.12
0	$V_{f0.5} - V_{\text{OPC}}$	40	+1.04	+0.75	+1.33
0	$V_{\text{adapt}} - V_{\text{OPC}}$	40	+1.08	+0.70	+1.46
0	$V_{\text{adapt}} - V_{f0.5}$	40	+0.04	-0.13	+0.21
0	H-nll_dr	10	+1.24	-1.28	+3.76
0.25	$V_{\text{OPC}} - V_{\text{NLL}}$	40	-1.26	-1.61	-0.91
0.25	$V_{f0.25} - V_{\text{NLL}}$	40	-0.26	-0.55	+0.03
0.25	$V_{f0.5} - V_{\text{NLL}}$	40	-0.10	-0.41	+0.20
0.25	$V_{f0.75} - V_{\text{NLL}}$	40	+0.12	-0.15	+0.39
0.25	$V_{\text{adapt}} - V_{\text{NLL}}$	40	+0.19	-0.06	+0.43
0.25	$V_{f0.5} - V_{\text{OPC}}$	40	+1.16	+0.84	+1.48
0.25	$V_{\text{adapt}} - V_{\text{OPC}}$	40	+1.45	+1.06	+1.83
0.25	$V_{\text{adapt}} - V_{f0.5}$	40	+0.29	+0.03	+0.54
0.5	$V_{\text{OPC}} - V_{\text{NLL}}$	40	-1.24	-1.59	-0.89
0.5	$V_{f0.25} - V_{\text{NLL}}$	36	+0.30	-0.09	+0.68
0.5	$V_{f0.5} - V_{\text{NLL}}$	40	+0.48	+0.14	+0.82
0.5	$V_{f0.75} - V_{\text{NLL}}$	36	+0.73	+0.39	+1.07
0.5	$V_{\text{adapt}} - V_{\text{NLL}}$	36	+0.73	+0.44	+1.03
0.5	$V_{f0.5} - V_{\text{OPC}}$	40	+1.72	+1.27	+2.16
0.5	$V_{\text{adapt}} - V_{\text{OPC}}$	36	+2.02	+1.60	+2.45
0.5	$V_{\text{adapt}} - V_{f0.5}$	36	+0.28	+0.04	+0.52
0.75	$V_{\text{OPC}} - V_{\text{NLL}}$	40	-1.10	-1.49	-0.70
0.75	$V_{f0.5} - V_{\text{NLL}}$	40	+1.11	+0.74	+1.49
0.75	$V_{f0.5} - V_{\text{OPC}}$	40	+2.21	+1.76	+2.66
1	$V_{\text{OPC}} - V_{\text{NLL}}$	40	-0.89	-1.33	-0.46
1	$V_{f0.25} - V_{\text{NLL}}$	34	+0.82	+0.36	+1.28
1	$V_{f0.5} - V_{\text{NLL}}$	40	+1.57	+1.10	+2.04
1	$V_{f0.75} - V_{\text{NLL}}$	34	+1.63	+1.20	+2.07
1	$V_{\text{adapt}} - V_{\text{NLL}}$	34	+1.59	+1.16	+2.03
1	$V_{f0.5} - V_{\text{OPC}}$	40	+2.46	+1.98	+2.95
1	$V_{\text{adapt}} - V_{\text{OPC}}$	34	+2.54	+2.07	+3.00
1	$V_{\text{adapt}} - V_{f0.5}$	34	+0.05	-0.27	+0.36
2	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+0.77	+0.14	+1.39
2	$V_{f0.25} - V_{\text{NLL}}$	40	+2.70	+2.01	+3.39
2	$V_{f0.5} - V_{\text{NLL}}$	40	+3.32	+2.74	+3.90
2	$V_{f0.75} - V_{\text{NLL}}$	40	+2.92	+2.42	+3.42
2	$V_{\text{adapt}} - V_{\text{NLL}}$	40	+2.97	+2.44	+3.50
2	$V_{f0.5} - V_{\text{OPC}}$	40	+2.55	+2.10	+3.01
2	$V_{\text{adapt}} - V_{\text{OPC}}$	40	+2.20	+1.71	+2.69
2	$V_{\text{adapt}} - V_{f0.5}$	40	-0.35	-0.71	+0.00
4	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+3.09	+2.41	+3.77
4	$V_{f0.5} - V_{\text{NLL}}$	40	+5.33	+4.70	+5.95
4	$V_{f0.5} - V_{\text{OPC}}$	40	+2.24	+1.67	+2.81
10	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+4.53	+3.69	+5.37
10	$V_{f0.5} - V_{\text{NLL}}$	40	+6.70	+5.78	+7.62
10	$V_{f0.5} - V_{\text{OPC}}$	40	+2.17	+1.35	+2.99
100	$V_{\text{OPC}} - V_{\text{NLL}}$	40	+4.73	+3.76	+5.69
100	$V_{f0.25} - V_{\text{NLL}}$	34	+6.09	+4.93	+7.24
100	$V_{f0.5} - V_{\text{NLL}}$	40	+6.90	+5.85	+7.95
100	$V_{f0.75} - V_{\text{NLL}}$	34	+6.13	+5.31	+6.96
100	$V_{\text{adapt}} - V_{\text{NLL}}$	34	+6.04	+5.16	+6.93
100	$V_{f0.5} - V_{\text{OPC}}$	40	+2.18	+1.40	+2.95
100	$V_{\text{adapt}} - V_{\text{OPC}}$	34	+1.36	+0.59	+2.12
100	$V_{\text{adapt}} - V_{f0.5}$	34	-0.59	-1.23	+0.05
100	H-nll_dr	10	+5.78	+3.79	+7.77

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.25	$V_{\text{OPC}} - V_{\text{NLL}}$	40	-0.04	-0.27	+0.19

continued: ml / SoftThreshτ1.5 DiD

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.25	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+0.28	+0.01	+0.55
0	0.25	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.08	-0.20	+0.36
0	0.25	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+0.24	+0.00	+0.48
0	0.25	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+0.33	+0.08	+0.58
0	0.25	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.12	-0.18	+0.42
0	0.25	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+0.37	+0.02	+0.71
0	0.25	$V_{\mathrm{adapt}} - V_{f0.5}$	40	+0.24	-0.03	+0.52
0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	-0.02	-0.32	+0.28
0	0.5	$V_{f0.25} - V_{\mathrm{NLL}}$	36	+0.82	+0.41	+1.24
0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+0.66	+0.25	+1.07
0	0.5	$V_{f0.75} - V_{\mathrm{NLL}}$	36	+0.86	+0.46	+1.27
0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	36	+0.87	+0.49	+1.25
0	0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+0.68	+0.30	+1.06
0	0.5	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	36	+0.91	+0.50	+1.32
0	0.5	$V_{\mathrm{adapt}} - V_{f0.5}$	36	+0.24	-0.02	+0.50
0	0.75	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.13	-0.29	+0.54
0	0.75	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.30	+0.87	+1.73
0	0.75	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.17	+0.77	+1.57
0	1	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+0.33	-0.16	+0.81
0	1	$V_{f0.25} - V_{\mathrm{NLL}}$	34	+1.26	+0.76	+1.77
0	1	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+1.75	+1.21	+2.30
0	1	$V_{f0.75} - V_{\mathrm{NLL}}$	34	+1.71	+1.19	+2.24
0	1	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	34	+1.68	+1.12	+2.24
0	1	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.42	+0.94	+1.91
0	1	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	34	+1.39	+0.86	+1.91
0	1	$V_{\mathrm{adapt}} - V_{f0.5}$	34	+0.03	-0.31	+0.37
0	2	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+1.99	+1.32	+2.65
0	2	$V_{f0.25} - V_{\mathrm{NLL}}$	40	+3.24	+2.49	+3.99
0	2	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+3.50	+2.86	+4.15
0	2	$V_{f0.75} - V_{\mathrm{NLL}}$	40	+3.04	+2.47	+3.61
0	2	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	40	+3.11	+2.50	+3.71
0	2	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.52	+1.02	+2.02
0	2	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	40	+1.12	+0.59	+1.66
0	2	$V_{\mathrm{adapt}} - V_{f0.5}$	40	-0.39	-0.79	-0.00
0	4	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+4.31	+3.60	+5.02
0	4	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+5.51	+4.82	+6.20
0	4	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.20	+0.61	+1.80
0	10	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+5.75	+4.91	+6.60
0	10	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+6.88	+5.97	+7.79
0	10	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.13	+0.36	+1.90
0	100	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	40	+5.95	+4.98	+6.91
0	100	$V_{f0.25} - V_{\mathrm{NLL}}$	34	+6.53	+5.43	+7.62
0	100	$V_{f0.5} - V_{\mathrm{NLL}}$	40	+7.09	+6.06	+8.12
0	100	$V_{f0.75} - V_{\mathrm{NLL}}$	34	+6.21	+5.41	+7.02
0	100	$V_{\mathrm{adapt}} - V_{\mathrm{NLL}}$	34	+6.13	+5.29	+6.97
0	100	$V_{f0.5} - V_{\mathrm{OPC}}$	40	+1.14	+0.34	+1.93
0	100	$V_{\mathrm{adapt}} - V_{\mathrm{OPC}}$	34	+0.21	-0.59	+1.00
0	100	$V_{\mathrm{adapt}} - V_{f0.5}$	34	-0.61	-1.24	+0.02
0	100	H-nll_dr	10	+4.54	+1.09	+7.99

6.13 myket / R*

Gammas present: $0, 0.5$. Seeds n vary by method/cell (see tables).

Absolute greedy CTR (%) by γ and method

γ	n	method	mean CTR%	95% CI lo	95% CI hi
0	20	OPC/DR	19.90	19.23	20.58
0	20	NLL	21.31	20.71	21.91
0	20	NLL→DR (ref)	18.63	15.92	21.35
0	20	hybrid f=0.5	21.39	20.81	21.98
0.5	20	OPC/DR	19.74	18.72	20.77
0.5	20	NLL	20.60	19.64	21.55
0.5	20	NLL→DR (ref)	19.48	17.41	21.54
0.5	20	hybrid f=0.5	21.37	20.33	22.41

Paired contrasts (CTR points) by γ

γ	contrast	n	mean	95% CI lo	95% CI hi
0	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-1.41	-2.00	-0.82
0	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.08	-0.28	+0.45
0	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+1.49	+0.91	+2.08
0	H-nll_dr	20	+2.76	+0.21	+5.32
0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	-0.85	-1.59	-0.12
0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.77	+0.29	+1.25
0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+1.62	+1.01	+2.24
0.5	H-nll_dr	20	+1.89	+0.14	+3.64

DiD vs γ=0 (CTR points) with 95% CI

Definition: DiD = $\Delta(\gamma_1) - \Delta(\gamma_0)$ for the named contrast, paired over seeds present at both γ.

γ_0	γ_1	contrast	n	DiD	95% CI lo	95% CI hi
0	0.5	$V_{\mathrm{OPC}} - V_{\mathrm{NLL}}$	20	+0.55	-0.20	+1.31
0	0.5	$V_{f0.5} - V_{\mathrm{NLL}}$	20	+0.69	+0.16	+1.22
0	0.5	$V_{f0.5} - V_{\mathrm{OPC}}$	20	+0.13	-0.54	+0.80
0	0.5	H-nll_dr	20	-0.87	-2.92	+1.18

7 Reproducibility

Tags live under `artifacts/full_study/run_*`. Analysis: `artifacts/full_study/misspec_experiment/analyze_hybrid_by_`
→ `hybrid_by_cell/*.csv`. This PDF: `build_hybrid_results_pdf.py`.